

SYSTEM OPERATION, TESTING & VALIDATION HANDBOOK

Trustworthy AI for Protein Engineering

Discovering Mutation-Space Reliability and Applying It to Industrial Carbon-Capture Enzyme Design

Version 2.0

Purpose Of This Handbook

This handbook is the operational guide for the entire project.

Unlike the Blueprint, which explains the science, or the Implementation Roadmap, which explains the development sequence, this handbook explains how to actually operate, test, validate, and troubleshoot the system.

This document answers the questions:

- What do I run?
- What should happen?
- How do I know it worked?
- What files should be produced?
- What outputs should I inspect?
- What constitutes passing validation?
- What should I do if results appear incorrect?

This handbook should be used continuously throughout development.

System Summary

The system follows the workflow:

Source Data



Sparsity Framework



Experiment 1
(Antisymmetry)



Experiment 2
(Epistasis)



Integrated Reliability Analysis



Reliability Framework



TKT Mutation Landscape



Industrial Fitness Framework



Publication Outputs

The goal of the system is to answer:

Can mutation-space sparsity
predict prediction failure?

and:

Can that information improve
protein-engineering decisions?

PHASE 1

Infrastructure Validation

Purpose

Validate that the software environment and project foundation are functional.

Expected Components

Configuration Framework

Logging Framework

Dataset Registry

Validation Framework

Experiment Tracking

Unit Test System

Validation Checklist

Verify:

Configuration loads successfully

``

Verify:

Logging generates output

Verify:

Dataset registry initializes

Verify:

Unit tests execute

Verify:

No startup exceptions occur

Pass Criteria

Application starts successfully

Tests pass

Logging works

Configuration loads

Failure Indicators

Missing configuration

Import errors

Registry initialization failure

Broken unit tests

PHASE 2

Dataset Validation

Purpose

Validate all source datasets before scientific analysis begins.

Required Source Files

megascale_d.parquet

protein_sequences.parquet

protein_structures.parquet

Validation Procedure

Verify:

Dataset loads successfully

Verify:

Schema matches specification

Verify:

Primary keys are unique

Verify:

No unexpected columns exist

Verify:

Required columns exist

,

Verify:

Missing values within tolerance

Pass Criteria

All datasets loaded

All schemas validated

All keys unique

Failure Indicators

Duplicate keys

Schema mismatch

Missing required fields

Corrupt files

PHASE 3

Sparsity Framework Validation

Purpose

Validate all sparsity calculations.

Expected Outputs

empirical_sparsity.parquet

evolutionary_sparsity.parquet

structural_sparsity.parquet

combined_sparsity.parquet

Empirical Sparsity Validation

Verify:

All mutations processed

Verify:

Frequencies normalized

Verify:

Output range between 0 and 1

Check:

Most common mutations

↓

Lowest sparsity

Evolutionary Sparsity Validation

Verify:

ESM probabilities generated

Verify:

Log probabilities calculated

Verify:

Values normalized

Verify range:

0.0 – 1.0

Structural Sparsity Validation

Verify:

SASA values generated

Verify:

Normalization successful

Verify:

No negative SASA values

Combined Sparsity Validation

Verify:

All source datasets merged

Verify:

Combined Sparsity exists

Verify:

No missing values

Verify:

$0.0 \leq \text{Combined Sparsity} \leq 1.0$

Pass Criteria

combined_sparsity.parquet created

No validation errors

All values normalized

PHASE 4

Experiment 1 Validation

Sparsity and Thermodynamic Inconsistency

Purpose

Validate antisymmetry calculations.

Expected Outputs

antisymmetry_predictions.parquet

antisymmetry_results.parquet

Validation Procedure

Verify:

Forward predictions exist

Verify:

Reverse predictions exist

Verify:

Antisymmetry Error calculated

Formula check:

Antisymmetry Error

=

| Forward + Reverse |
``

Randomly inspect records manually.

Distribution Validation

Expected:

Most records
near low error values

Investigate:

Extreme outliers

Figure Validation

Required figure:

Combined Sparsity

vs

Antisymmetry Error

Verify:

Axes labeled

Data points present

Figure exported correctly

Pass Criteria

Antisymmetry dataset created

Figures generated

Manual checks pass

PHASE 5

Experiment 2 Validation

Sparsity and Epistasis Error

Purpose

Validate prediction-failure calculations.

Expected Outputs

epistasis_results.parquet

Validation Procedure

Verify:

Experimental epistasis exists

Verify:

Predicted epistasis exists

Verify:

Epistasis Error calculated

Formula check:

Error

=

| Predicted – Experimental |

Perform random manual verification.

Figure Validation

Required figure:

Combined Sparsity

vs

Epistasis Error

Verify:

Data plotted correctly

No missing values

Export successful

Pass Criteria

Epistasis dataset generated

Figures generated

Validation checks passed

PHASE 6

Integrated Reliability Analysis

Purpose

Validate the core scientific relationship.

Expected Outputs

integrated_reliability_analysis.parquet

Required Relationships

Analysis 1

Sparsity

↓

Antisymmetry Error

Analysis 2

Sparsity

↓

Epistasis Error

Analysis 3

Antisymmetry Error

↓

Epistasis Error

Validation Procedure

Verify:

Integrated dataset built

Verify:

No missing fields

Verify:

All three relationships analyzed

Primary Figure

Sparsity



Antisymmetry



Epistasis

This becomes a candidate figure for publication and conference presentations.

Pass Criteria

Integrated analysis complete

Figure generated

Statistics exported

PHASE 7

Reliability Framework Validation

Purpose

Validate reliability calculations.

Expected Output

reliability_scores.parquet

Calculation Check

Formula:

Reliability

=

1 – Combined Sparsity

Randomly verify records manually.

Category Validation

Expected categories:

High

Moderate

Low

Very Low

Verify:

Every score classified exactly once

Range Validation

Verify:

$0.0 \leq \text{Reliability} \leq 1.0$

Pass Criteria

Scores valid

Categories valid

Dataset complete

PHASE 8

TKT Mutation Landscape Validation

Purpose

Validate full mutation-space generation.

Expected Outputs

tkt_single_mutation_landscape.parquet

tkt_mutation_analysis.parquet

Mutation Count Validation

Calculate:

Protein Length

×

19

Expected result:

Total Number Of Mutations

Verify dataset row count matches expectation.

Mutation Integrity Validation

Verify:

No duplicate mutations

No missing positions

All substitutions generated

Reliability Integration Validation

Verify:

Combined Sparsity exists

Reliability exists

Stability exists

for every mutation.

Pass Criteria

Entire TKT landscape generated

Expected row count achieved

No duplicate entries

PHASE 9

Active-Site Analysis Validation

Purpose

Validate control analysis.

Expected Output

active_site_analysis.parquet

Validation Procedure

Verify:

Active-site residues defined

Verify:

Distances calculated

Verify:

No negative distances

Verify:

Summary statistics generated

Pass Criteria

Distance calculations verified

Dataset created

PHASE 10

Industrial Fitness Framework Validation

Purpose

Validate engineering ranking system.

Expected Outputs

industrial_fitness_scores.parquet

top_candidate_mutations.csv

Score Validation

Verify:

Predicted Stability present

Reliability present

Industrial Fitness present

Verify:

Scores normalized

Ranking Validation

Verify:

Sorted descending

Verify:

No rank duplication

Verify:

Rank 1 > Rank 2 > Rank 3

according to score.

Pass Criteria

Rankings generated

Scores valid

Candidate list produced

FIGURE VALIDATION PROCEDURES

Discovery Figures

Verify:

Combined Sparsity

vs

Antisymmetry Error

Contains:

Title

Axes

Data

Exported PNG

Validation Figures

Verify:

Combined Sparsity

vs

Epistasis Error

Reliability Figures

Verify:

Reliability Distribution

Reliability Categories

TKT Figures

Verify:

Mutation Landscape

Reliability Landscape

Industrial Fitness Figures

Verify:

Top Ranked Candidates

Fitness Distribution

KILLER FIGURE VALIDATION

Required Components

The Killer Figure must contain:

Megascale-D

↓

Combined Sparsity

↓

Antisymmetry Error

↓

Epistasis Error

↓

Reliability Framework

↓

Complete TKT Landscape

↓

Industrial Fitness Ranking

↓

Top Candidate Mutations

Validation Criteria

A reviewer should be able to understand:

Discovery

Validation

Application

within approximately ten seconds.

The figure should communicate the complete scientific narrative without requiring explanation.

FINAL PROJECT VALIDATION CHECKLIST

Infrastructure



Datasets



Empirical Sparsity



Evolutionary Sparsity



Structural Sparsity



Combined Sparsity



Experiment 1



Experiment 2



Integrated Reliability Analysis



Reliability Framework



TKT Mutation Landscape



Active-Site Analysis



Industrial Fitness Framework



Candidate Ranking



Publication Figures



Killer Figure



Project Completion Definition

The project is considered complete when all datasets, analyses, figures, and validation procedures described in this handbook have been successfully executed and verified.

At that point, the system should be capable of answering:

Does mutation-space sparsity
predict prediction failure?
Can prediction reliability
be estimated?
Can reliability improve
protein-engineering decisions?

through a fully reproducible computational framework for trustworthy AI-guided protein engineering.

End Of System Operation, Testing & Validation Handbook

At this point, the core document stack is complete. My recommendation before implementation begins is to create one additional final document:

PROJECT MASTER INDEX

This document would map:

All Documents

All Phases

All Datasets

All Figures

All Outputs

All Cursor Prompts

All Validation Steps

into a single navigation guide. For a project of this size, that index becomes extremely valuable during development, paper writing, science fair preparation, and ISEF presentation preparation.